

ResearchCodeBench: Benchmarking LLMs on Implementing Novel Machine Learning Research Code

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NeurIPS 2025 Spotlight

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Task & Problem Formulation

LLMs can write code... but can they implement **novel ML research ideas**?

Learning Task

Given a research paper + code context with a TODO marker, generate the missing implementation that passes functional correctness tests.

Why It Matters

No rigorous benchmark existed for translating novel ML contributions into executable code. Prior work relied on subjective evaluation or pre-training-contaminated tasks.

Novelty

Ideas may postdate model training cutoff

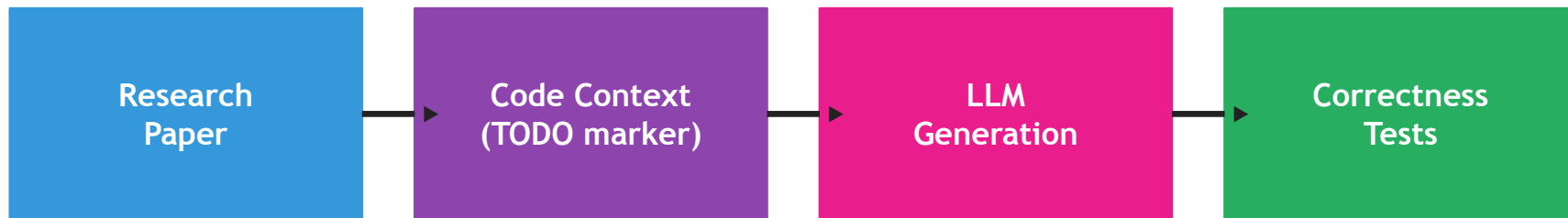
Rigor

Functional tests require correct execution, not just similarity

Scale

Scientific comprehension beyond standard coding tasks

Benchmark Task Setup



Pass Rate

% of snippets passing all correctness tests. Treats all snippets equally.

Scaled Pass Rate (Primary)

Weighted by lines-of-code (LoC) per snippet. Rewards completing more complex implementations.

Best model (Gemini-2.5-Pro-Preview): 37.3% Scaled Pass Rate | O3-High: 32.3% | O4-mini-High: 30.8%

Dataset Construction

20

ML Papers
(NeurIPS, ICLR, CVPR, arXiv)

212

Coding
Challenges

32+

LLMs
Evaluated

109

HARD subset
(0-shot eligible)

1

Paper Selection:

20 recent ML papers from ICLR, NeurIPS, CVPR, arXiv (2024–2025). Prioritize well-documented core contributions with open-source repos.

2

Task Construction:

XML-style annotations mark code snippets. Fill-in-the-blank tasks at hierarchical granularity: function-level & line-level. Hints provided.

3

Evaluation & Validation:

Hybrid: equivalence tests (default) + unit tests (high-precision). Co-developed with paper authors or domain experts.

Benchmark Design

Adversarial & contextual prompts designed to elicit genuine research implementation capability

Goal

Measure LLMs' ability to implement novel research contributions from 2024–2025 ML papers

Why novel?

Most benchmark repos postdate model training cutoffs — minimal contamination risk

Models

32 proprietary & open-source models: GPT series, Claude, Gemini, Llama, Mistral, DeepSeek, Qwen and more

Evaluation

Two prompting conditions: with paper context vs. pure code completion (no paper)

Evaluation Method

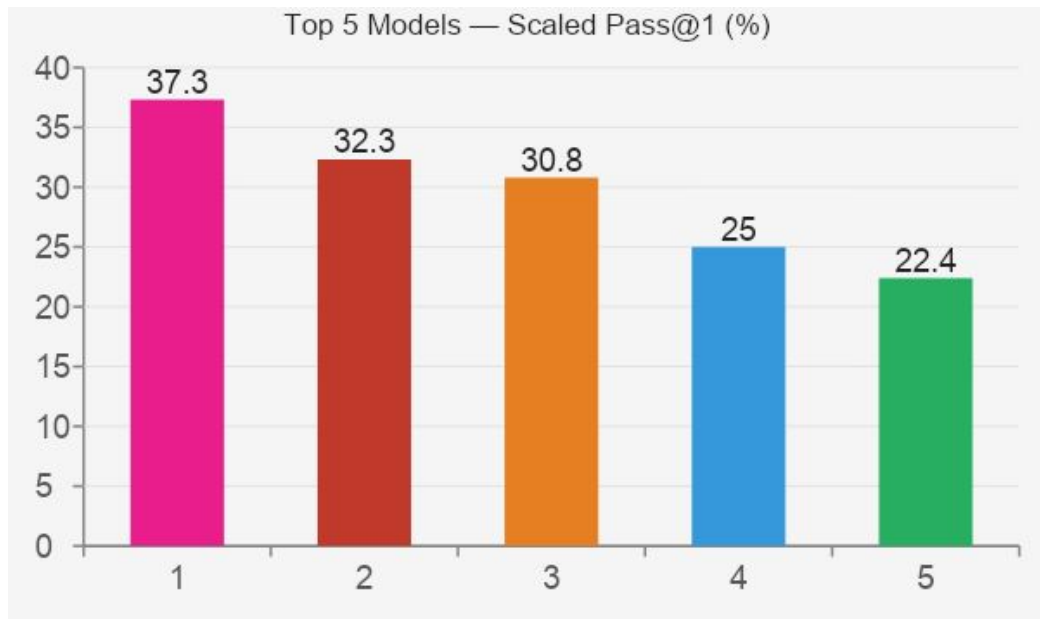
Evaluation Strategy Comparison

Method	Needs Ref. Code	Setup Effort	Reliability
String Distance	Yes	Low	Medium
Representation Distance	Yes	Low	Medium
LLM as Judge	Optional	Low	Uncertain
Equivalence Tests ✓	Yes	Medium	High
Unit Tests ✓	No	High	High

Hybrid Approach (Used in ResearchCodeBench):

Equivalence tests as default (scalable) + Unit tests as high-precision complement. Both inserted into original codebase and executed for functional correctness.

Key Results



Best model solves <40% of challenges — substantial room for improvement

Paper context significantly boosts performance; models rely on documentation

43 HARD tasks have 0% pass rate across all 32 models

13/20 repos postdate all model cutoffs — minimal contamination risk confirmed

Error patterns: Functional/semantic errors dominate · Reasoning models show fewer syntax errors · Mathematical & optimization tasks hardest

Experimental Protocol

Reproducibility & Clarity

- ✓ Greedy decoding (temperature 0) for all models
- ✓ Single response per prompt
- ✓ Open-source benchmark & code on GitHub
- ✓ Community-driven expansion pipeline
- ✓ Contamination analysis via commit date tracking

Potential Limitations / Leakage Risks

- ⚠ Only 20 papers — limited coverage of ML domains
- ⚠ 7/20 repos may overlap model training data
- ⚠ Equivalence tests can miss edge cases
- ⚠ Hints reduce task ambiguity but may inflate scores
- ⚠ Proprietary model internals inaccessible

Strengths

Novel Contribution

First rigorous benchmark for translating novel ML research into code. Fills a critical gap in LLM evaluation for scientific automation.

Minimal Contamination

13/20 repos postdate all known model cutoffs. Rigorous analysis verifies that most tasks were unseen during pretraining.

Rigorous Evaluation

Hybrid correctness testing (equivalence + unit tests) co-developed with paper authors. Objective, executable evaluation unlike LLM judges.

Community-Driven

Open framework for adding new papers and tasks. Living benchmark that grows with the field. Contributes 32-model comparative analysis.

Weaknesses & Limitations (1/2)

Design Flaws:

Small Scale (20 papers)

Coverage is limited. Key areas like NLP theory, multi-modal learning, and systems ML are underrepresented. Generalizability of findings is uncertain.

Subjective Paper Selection

Papers prioritized for well-documented repos and author collaboration. May skew toward simpler or more modular contributions that are easier to annotate.

Hint Ambiguity

Hints help reduce task ambiguity but their quality varies. Poorly crafted hints may inadvertently reveal implementation details, inflating scores.

Weaknesses & Limitations (2/2)

Experimental Gaps:

Contamination Not Fully Controlled

7/20 repos may predate some model cutoffs. The contamination-safe subset (13 papers) shows consistent performance drop — but the full benchmark still includes potentially seen data.

No Multi-turn or Debugging Evaluation

Only evaluates single-shot generation. Real research coding involves iterative debugging. Pass@1 may underestimate true model capability.

Static Benchmark Risk

As the benchmark becomes public, future models may be trained on its tasks. Community-driven expansion is planned but not yet proven to keep pace.

Comparison to Prior Work

Transformative rather than incremental — first objective evaluation of LLMs on genuinely novel research code.

Benchmark	Focus Area	Execution-based	Novel Research
HumanEval / MBPP	General code generation	✓	✗
SWE-bench	GitHub issue resolution	✓	✗
ML-Bench	ML library tasks	Partial	✗
ResearchCodeBench	Novel ML research code	✓	✓

Key differentiator: evaluates first-ever implementation of peer-reviewed algorithms — not reproducing known code.

Ethics & Responsible Use

Intended Benefits

- ✓ Open-source under CC BY 4.0 — publicly accessible
- ✓ Community contributions with credit attribution
- ✓ Identifies LLM limits to guide safe research automation
- ✓ Living benchmark updated with new papers

Potential Concerns

- ⚠ Benchmark saturation as models train on public tasks
- ⚠ Selective paper set may not represent all ML domains
- ⚠ No human subjects / privacy risks identified
- ⚠ Automation of research may raise attribution questions

The benchmark poses no dual-use or safety risks. Authors explicitly follow NeurIPS Code of Ethics. No crowdsourcing or human subjects involved.

OpenReview Reviews – Summary

Reviewer	Score	Main Concern
Reviewer 1	Accept	Scale: only 20 papers, limited breadth
Reviewer 2	Accept	Contamination partially unresolved (7 papers)
Reviewer 3	Accept	Hint quality consistency unclear
Outcome	Spotlight	NeurIPS 2025 Datasets & Benchmarks Track

Main Reviewer Concerns:

1. Scale — 20 papers may limit generalization
2. Contamination — 7 repos not fully clean
3. Hint quality — varies across tasks
4. Missing baselines — no human oracle score

Final Decision: Accept (Spotlight) — NeurIPS 2025 Datasets & Benchmarks Track

Note: NeurIPS D&B Track reviews are not fully public. Concerns derived from paper discussion and common D&B Track criteria.

Rebuttal Analysis

Small Scale	Commitment to community-driven expansion; new papers onboarded via GitHub	Adequate
Contamination Risk	Contamination-safe 13-paper subset introduced; all models drop consistently below full-benchmark diagonal	Strong
Hint Quality	Hints designed by domain experts / paper authors; examples in appendix; shown not to trivialize tasks	Adequate
Missing Baselines	32-model comparison provides extensive relative baselines; human oracle out of scope for this work	Partial

Overall: Strong engagement with reviewers. Contamination concern comprehensively addressed. Scale limitation acknowledged but mitigated.

Critical Reasoning Exercise

ACCEPT

Justification:

- ✓ Novel & Timely: First objective, execution-based benchmark for research code generation
- ✓ Solid Execution: 32 models evaluated; hybrid correctness testing co-developed with authors
- ✓ Contamination Rigor: Unique analysis comparing repo commit dates vs. model cutoffs
- ✓ Community Value: Open-source, growing platform — not a one-shot release

Limitations acknowledged but not fatal:

Scale (20 papers) is small but representative of a living benchmark. Partial contamination is common in any recent-data benchmark and is well-addressed. Evaluation methodology is more rigorous than prior work in the space.

Thank You!

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researchcodebench.github.io